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Roll No

MI-601 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Mining Environment - II
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Derive the expression for theoretical head of centrifugal fan?
2. Explain in detail about ascensional and descensional ventilation. Which one is more preferred in coal mines and why? Supplement your answer with neat diagrams.
3. Explain the characteristics curve of fan in series and parallel, operated with the main mine fan.
4. Discuss the working of Gravimetric dust sampler MRE113A along with diagram.
5. Briefly explain:
 - i) Erases
 - ii) Booster fans
6. Explain :
 - a) Purpose of evasion
 - b) Laws of mine air friction
 - c) Equivalent orifice of a mine
 - d) Theoretical depression of a fan
 - e) Mechanical efficiency
7. Give the classification and physiological effects of dust.
8. Write short notes on:(Any four)
 - a) Ascensional and Descensional ventilation
 - b) Homotropical and Antitropical ventilation
 - c) Air crossing
 - d) Regulator
 - e) Splitting of air current
